

VisionAid STATS Developer Guide

Georgia Tech OMSCS C4G, Spring 2026

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PREVIOUS DOCUMENTATION

Spring 2025 Developer Documentation Link:

- <https://github.gatech.edu/cs-6150-computing-for-good/va-stats/blob/staging/public/documentation/Developer%20Guide.pdf>

PROJECT DESCRIPTION

The VisionAid STATS project by VisionAid Academy is a web-based application designed to assist the VisionAid team manage its training programs for visually impaired students.

The application allows staff to manage courses offered by the program, batches for courses, and student enrollment, attendance, and grades for each batch. The system has role-based access built in, granting different permissions to features and data for administrators, program managers, staff, and trainers.

One major focus of the application is accessibility. Keyboard integration, screen reader compatibility, and a user-friendly interface enable users to access and manage the training programs and students.

The platform also provides dashboards for quick insights, supports tracking student progress through assessments, and includes features to promote improved program and data management tasks.

ARCHITECTURE DIAGRAM

The VisionAid STATS application has a React + Next.JS frontend and a MySQL DB backend.

It is deployed to AWS public cloud via Vercel and leverages Google Cloud Platform (GCP) Google Auth0 for authentication.

As shown in the architecture diagram, the Vercel project and the GitHub repo are integrated such that GitHub automatically creates a preview deployment to test that there are no deployment errors upon every PR that is made to the GitHub repo.

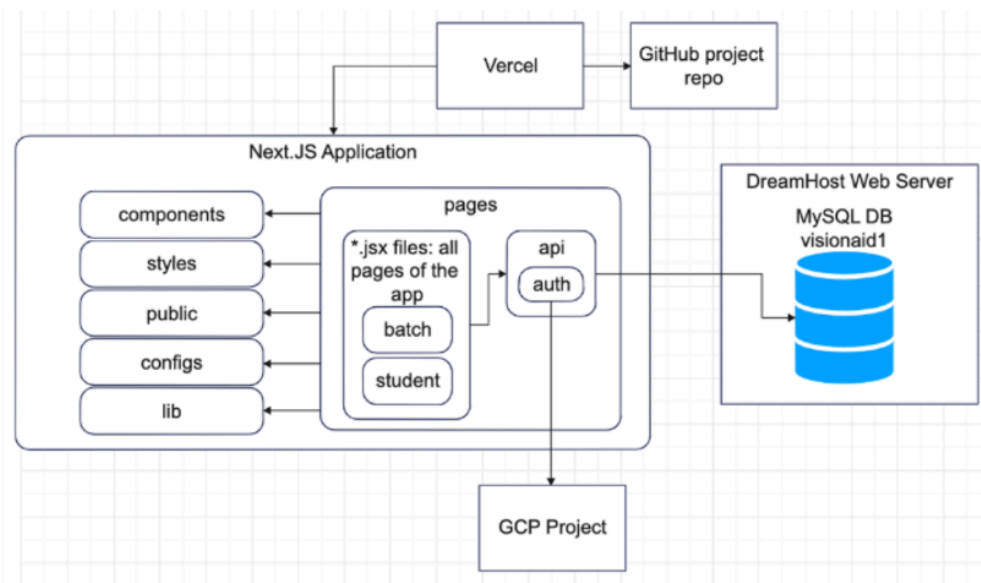
Vercel is also configured with environment variables specifying the credentials for the DB, so that the ``api`` module of the application can access it, as well as the GCP project

`GOOGLE\_CLIENT\_ID` and `GOOGLE\_CLIENT\_SECRET` so that the application's `auth` module can verify authenticated users stored in the project.

All application logic resides in the `pages` directory, which includes all *.jsx files that render the pages of the app.

Each page file includes api calls through methods defined in files in the `api` directory, which connect to the MySQL DB, make queries, and return results to the page.

The page files also leverage various helper scripts, reusable components (ie: Buttons, NavBar), icons, and CSS styles from the `components`, `styles`, `public`, `configs`, and `lib` directories as shown in the architecture diagram.



USER STORIES / DATA FLOWS

Here are the data flows for some of the key existing user stories:

User Stories	Data Flow
Add new staff member to system	1. Management user enters info for new staff member in the "Add New Staff Member" form 2. Form submission is handled in pages/users.jsx with a POST API call made to /api/usercreate 3. pages/api/usercreate.js takes form data, connects to visionaid1 database, and creates a new record in the vausers table with the data 4. users.jsx re-renders the page to fetch all the data with from

	vausers by making an API call to pages/api/getusersdata.js to show the newly entered data
Create new course	1. Management user enters info for new course in the “Create Course” form on the Courses page 2. Form submission is handled in pages/courses.jsx with a POST API call made to /api/coursecreate 3. pages/api/coursecreate.js takes form data, connects to visionaid1 database, and creates a new record in the vacourses table with the data 4. courses.jsx re-renders the page to fetch all the data from vacourses by making an API call to pages/api/getcoursedata.js to show the newly entered data
Create new batch for existing course	1. Management user enters info for new batch in the “Create Batch” form on the Batches page 2. Form submission is handled in pages/batches.jsx with a POST API call made to /api/batchcreate 3. pages/api/batchcreate.js takes form data, connects to visionaid1 database, and creates a new record in the vabatches table with the data 4. batches.jsx re-renders the page to fetch all the data from vabatches by making an API call to pages/api/getbatchdata.js to show the newly entered data
Register student in system	1. Management user enters info for new student in the form on the Student Registration page (under Students) 2. Form submission is handled in pages/studentregistration.jsx with a POST API call made to /api/studentapplication 3. pages/api/studentapplication.js takes form data, connects to visionaid1 database, and creates a new record in the vastudents table with the data 4. User navigates to Students page 5. students.jsx renders the page to fetch all the data from vastudents by making an API call to pages/api/getstudentsdata.js to show the newly entered data
Enroll a student in a batch	1. PM clicks on the “Roster” button from the table in the Batches page which redirects them to api/batch/[batch id] 2. On this page, they click the “Add Student” button which prompts them to select from a list of “Unassigned Students” to enroll them in the batch. The Unassigned Students list is fetched with the fetchUnassignedStudents method in pages/batch/[id].jsx and makes a POST API call to

	api/getunassignedstudents with a request body including the batch_id. The response is a list of student id's and names from the vastudents table where the student id is not in the list of students in the vastudent_to_batch table that are already enrolled in the batch of that batch id. 3. The addStudent method in pages/batch/[id].jsx then is called taking the studentId of the unassigned student to be added as input and makes a POST API call to api/addstudenttobatch where a record gets created in the vastudent_to_batch table with the student id and batch id. New records are also created in the va_grades table assigning all the existing assignment names for the batch to the student. New records are also inserted in va_attendance with the student id, batch id, and all the dates for the batch with the default being set to false. 4. Users should see a new row for students in the Grades and Attendance tables on the Batch Roster page.
Enter attendance for a batch	1. User toggles from "Absent" to "Present" for attendance for a student on a particular date in the attendance table 2. updateAttendance method in pages/batch/[id].jsx is called with the batchId, studentId, date, and the new isPresent value as input. The method makes a POST API call to api/updateattendance with the request body including all the method inputs. The corresponding record in va_attendance gets updated with the new isPresent value. 3. Attendance table in the batch's roster page re-renders to show the updated value
Add new assignment for a batch	1. The user clicks the "Add Assignment" button in a batch's roster page and fills out a form with the name of the assignment. 2. addAssignment method in pages/batch/[id].jsx is called which makes a POST API call to api/addassignment with the assignmentName and batchId in the request body. In api/addassignment.js, all students belonging to the batch are fetched and for each of them, a new record is made in va_grades with the new assignment name and default grade of 0. 3. addAssignment methods re-fetches grades data to refresh Grades table with new records. A new column appears for the grades table with a new assignment.
View student enrollment history	1. The user clicks on the "Enrollment History" button on any row of the table in the Students page. 2. User is redirected to an Enrollment History page for the

	selected student which is rendered from <code>pages/student/[id].jsx</code> 3. <code>getStudentData</code> method is called which makes a POST API call to <code>api/getstudentdetails</code> with the request body including the <code>studentID</code>. A SQL query is made to fetch batch details of all the batches the student is enrolled in from a JOIN of the <code>vabatches</code> and <code>vastudent_to_batch</code> tables on the batch id. These batches are returned and presented as a table on the Enrollment History page for the student. NOTE: No updates can be made to any of the data presented in the page since it's meant to be a report of the student's enrollment history.
Cancel Attendance for a class on a given date	1. User clicks on "Batches" in the Navbar 2. User selects a batch on in the Actions column 3. Click on the button "Batch Attendance" 4. Find the date of the course to be cancelled and click the  button. It will ask if you want to cancel for all students; if so, click Ok. 5. The label "Cancelled" will appear for all students under that class date column.
Bulk Register students	1. Click on the "Bulk Registration" link in the Navbar. 2. Click the "+ Add Row" button to add rows to the table 3. Enter the student information for each column 4. Click the "Check Rows" button to validate the data in each cell. a. If errors exist, it will display the number of them in the Row Status column, and the cells with errors will gain a red border. b. If the cells in a row are valid, it will turn the row blue and say the row is ready to be registered. 5. Click on the "Register Valid Rows" to register (i.e., add to the database) any rows that are blue/validated. This will turn those rows green and they can be verified by checking the "Students" table.

RECOMMENDED HOSTING

- **Database:**
 - **DreamHost MySQL DB server.**
 - To administer the database (ie: make modifications to tables/schemas/manually manage records) go to

<https://west1.phpmyadmin.dreamhost.com/signon.php> and enter the login details (shared with Isaac W.).

- Follow phpMyAdmin documentation for guidelines on using the portal for database management.
- The same database can be used for both development and production environments.
- Use of DreamHost is free and requires proof of 501(c)(3) status.
- **Website:**
 - **Vercel deployment.**
 - Log in to the VA team's Vercel account and navigate to the Project page of the visionaid-stats-ng project to view the status of the Production Deployment.
 - Make sure the "STATUS" field is "Ready" and the latest commit message shown under "BRANCH" reflects your understanding of the latest commit of the main branch of the connected github repo.
 - Use of a Hobby account of Vercel is free. There is no foreseeable requirement to upgrade to the Pro account since the primary value of the Pro account is if multiple developers have their own Vercel accounts and want to work together in a Vercel Team as well as performance enhancements and available plugins which are not necessary to run the application.

APPLICATION INSTALLATION

DEV ENV Setup

1. Clone the GitHub repository
2. Add all necessary environment variables in the .env file in the project root directory. See the ENVIRONMENT_VARIABLES section below for details.
3. Get the database backup file from /db_backups directory and use it to set up the MySQL database using phpMyAdmin as instructed in the RECOMMENDED HOSTING section.
4. Add yourself as a user with role type MANAGEMENT in the vausers table using your gmail address which will be used for authentication.
5. Open the repo with Visual Studio Code.
6. In the repo, ensure all apiUrlEndpoint constants are set to paths of the form 'http://localhost:3000/...'
7. In the terminal from the project root directory, run 'npm run dev'
8. On a web browser navigate to <http://localhost:3000>

PRODUCTION ENV Setup

1. Ensure GitHub repository has Vercel integration specifying the link to the Vercel project in the GitHub project settings.
2. Ensure all apiUrlEndpoint constants are set to paths of the form ``https://visionaid-statsng.vercel.app/...``
3. Push changes to the “main” branch of the GitHub repo by merging PRs (Pull Requests) made to the repo or pushing changes directly to the “main” branch.
4. Check the deployment status on the Vercel project deployment page. If the deployment fails, logs will be shown on Vercel by clicking on the failed deployment. Follow the guidance in the error messages to resolve errors. Push fixes for the errors and ensure the deployment goes through smoothly.
5. Navigate to the project website <https://visionaid-stats-ng.vercel.app> to see the newly pushed changes.

ENVIRONMENT VARIABLES

- For local development, in the .env file from the root directory of the project repository, add the following environment variables:
 - MYSQL_HOST: MYSQL DB server hostname (ie: mysql.foo.dreamhosters.com)
 - MYSQL_DATABASE: name of database (visionaid1)
 - MYSQL_USER: database admin username
 - MYSQL_PASSWORD: database admin password
 - JWT_SECRET: generate a JWT token using Postman or command-line JWT token generator
 - GOOGLE_CLIENT_ID from GCP project with OAuth Consent setup
 - GOOGLE_CLIENT_SECRET from GCP project with OAuth Consent setup
- For production deployment on Vercel, ensure that the same environment variables are set in the project Settings.
- Ensure the .env file is in the .gitignore file so they are never pushed to the shared repo and always live locally on your machine.

AUTHENTICATION & AUTHORIZATION

Authentication with Google OAuth

1. Navigate to cloud.google.com
2. Menubar dropdown menu on the immediate right of 'Google Cloud'
3. Top right of dialog window: 'NEW PROJECT'
4. Enter 'Project name' and 'Location' if applicable and click 'CREATE' button
5. Click 'Google Cloud' in menubar and <YourProjectName> from the dropdown
6. Click 'DASHBOARD,' then 'APIs & Services'
7. Click 'Credentials,' then 'OAuth consent screen' in side menu
8. Select 'External' in the main panel and click 'CREATE'

9. Complete the 'App information' section in the main panel
10. Enter the protected views in the app, e.g., Students, Courses routes and click 'Create'
11. 'OAuth consent screen' in side menu
12. In 'Test users' section of main panel click 'ADD USERS' button
13. Add Gmail addresses for those that will test the app, and save
14. New user(s) now appear in the main panel

In `/pages/api/auth/[...nextauth].js`, the environment variables `GOOGLE_CLIENT_ID`, `GOOGLE_CLIENT_SECRET`, and `JWT_SECRET` are set for the Next.JS next-auth library to connect with the Google OAuth provider set up in earlier steps specified above.

Note: If a user who is not on the test users list tries to log in, they will receive a “not authorized” error. Make sure to add your own Gmail to test login during development.

Authorization: RBAC (Role-Based Access Control)

All of the files in the pages directory include a check on the signed-in users' role to ensure they are authorized to view the contents of the page. Some of the pages such as the Batches page have partial redactions depending on the role. For example, the Batch Creation form is not visible or accessible to users with the role `ADMINISTRATOR`, but the table of batches is visible to all signed-in users but none of the rows are editable or delete-able by users with non-PM or non-MANAGEMENT roles. Here is the full set of authorization rules implemented in the application for Management (MGMT), Program Managers (PM), and Administrators (ADMIN.):

PAGE	RESOURCE	ACTION	MGMT.	PM	ADMIN.	UNAUTH
Home	All	View				
Home	Login Button	Authenticate				
Home	Logout	Authenticate				
Batches	Batch	View				
Batches	Batch	Create				
Batches	Batch	Update				
Batches	Batch	Delete				
Students	Student	Add				
Students	Student	Remove				
Students	Student	Update				
Batch Details	Batch Details	View				
Users	User list	View				
Users	User	Create				
Users	User	Disable				
Batch Details	Batch Roster	View				
Batch Details	Batch Roster	Add student				
Batch Details	Batch Roster	Remove student				
Batch Details	Batch Roster	Update Attendance				
Courses	Course	Create				
Courses	Courses	Update				
Courses	Courses	Delete				
Students	Student	Form				
Batch Details	Batch Roster	Update Grades				

DATABASE

Database Backup

- Database backup: provided in the project repo in the /db_backups directory

Database Schemas

The visionaid1 database's table schemas can be viewed from the phpMyAdmin console. For example, to view the vausers table schema, navigate to the vausers table from the side panel and go to the "Structure" tab to view the schema, shown in the screenshot

below.

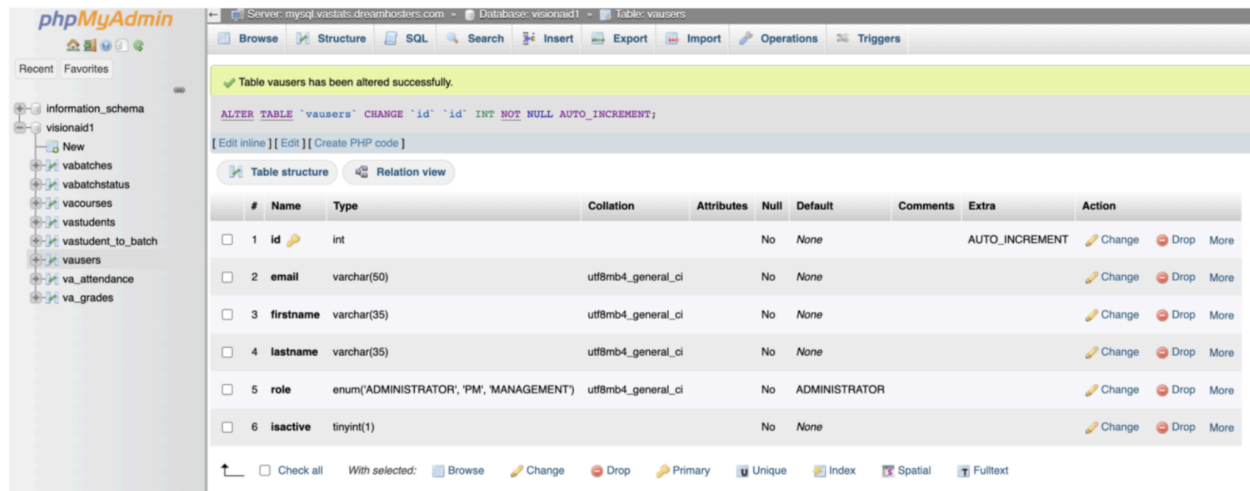


Table vausers has been altered successfully.

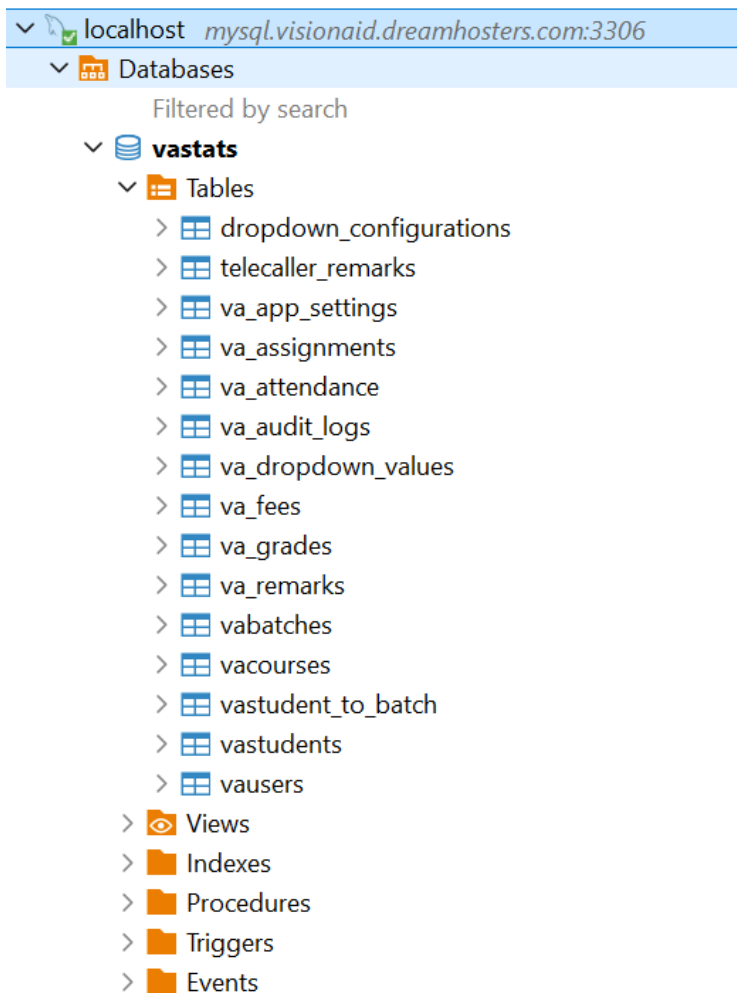
```
ALTER TABLE `vausers` CHANGE `id` `id` INT NOT NULL AUTO_INCREMENT;
```

[Edit inline] [Edit] [Create PHP code]

Table structure Relation view

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 id	int			No	None		AUTO_INCREMENT	Change Drop More
☐	2 email	varchar(50)	utf8mb4_general_ci		No	None			Change Drop More
☐	3 firstname	varchar(35)	utf8mb4_general_ci		No	None			Change Drop More
☐	4 lastname	varchar(35)	utf8mb4_general_ci		No	None			Change Drop More
☐	5 role	enum('ADMINISTRATOR', 'PM', 'MANAGEMENT')	utf8mb4_general_ci		No	ADMINISTRATOR			Change Drop More
☐	6 isactive	tinyint(1)			No	None			Change Drop More

☐ Check all With selected: [Browse](#) [Change](#) [Drop](#) [Primary](#) [Unique](#) [Index](#) [Spatial](#) [Fulltext](#)



localhost mysql.visionaid.dreamhosters.com:3306

Databases

Filtered by search

- vastats**
 - Tables
 - dropdown_configurations
 - telecaller_remarks
 - va_app_settings
 - va_assignments
 - va_attendance
 - va_audit_logs
 - va_dropdown_values
 - va_fees
 - va_grades
 - va_remarks
 - vabatches
 - vacourses
 - vastudent_to_batch
 - vastudents
 - vausers
 - Views
 - Indexes
 - Procedures
 - Triggers
 - Events

Database Schema Diagram

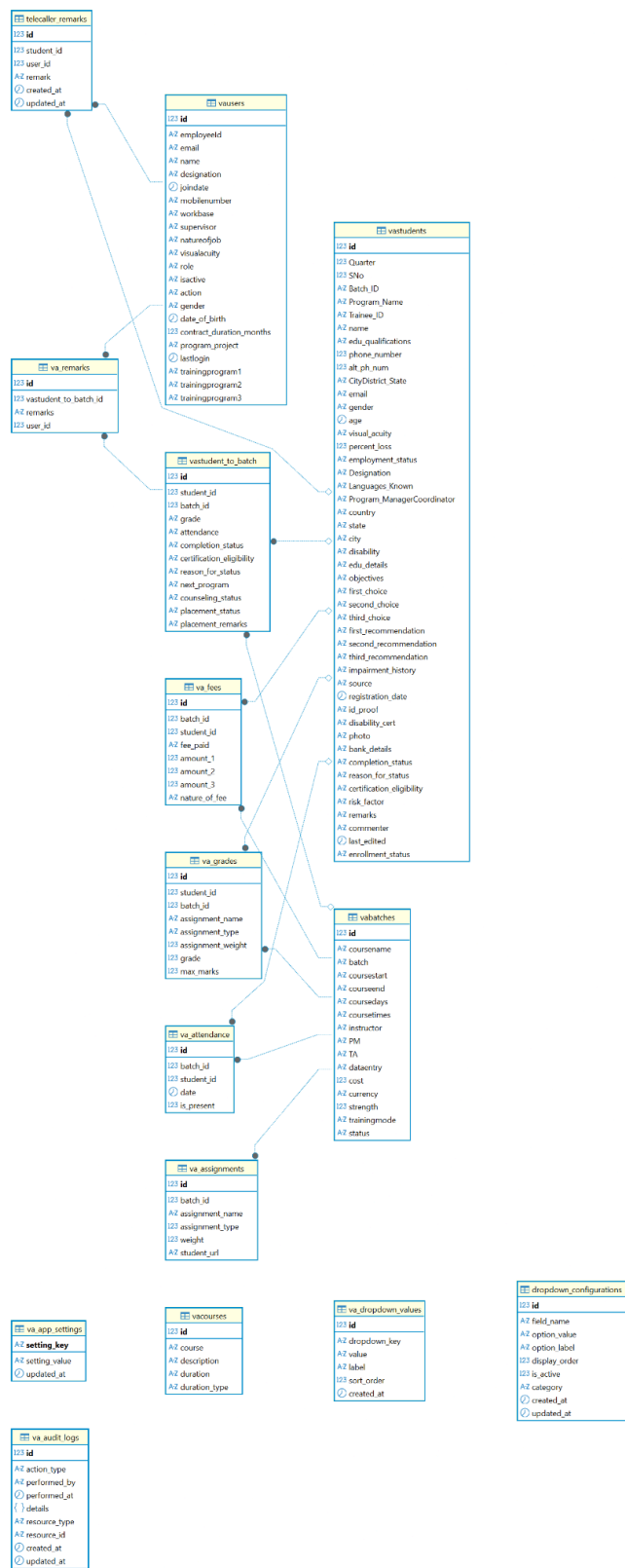


TABLE COMPONENT OVERVIEW

We use the Table component (/components/Tables.jsx) to display all tables on the website, such as the table of batches, students, users, and attendance/grades tables for batch rosters.

To utilize the table component, instantiate it with the constructor such as

```
<Table columns={coursesColumn} tableData={dataResponse} isDelete={isDelete} onDeleteClick={handleDeleteCourse}
isEditable={isEditable} onEditSave={handleUpdateCourse} Title={'Courses List'} />
```

Ensure the column headers are defined for the columns attribute, specifying at least the name (displayed as column header) and accessor (column name in dataset returned) for each column. Other properties for a column header element includes:

- column.width, which is the width of the column displayed on the webpage as a percentage of the total width
- column.type which is the type for the values in that column. For example, if the type is `enum` column.type should be set to `enum`.
- column.availableValues is an array of possible values if the column.type is `enum`. tableData is the dataset to be displayed as a table on the webpage.

'isDelete' and 'isEditable' should be set to true if the rows of the table should be editable and deletable. If they are, the onDeleteClick and onEditSave functions should be defined and specified as input for the Table.

Table component's helper functions are defined in the /utils/tableHelper.js file.

PWA (Progressive Web Application)

This project leverages PWA to make the application mobile-friendly. PWA is enabled on every website page by including a Next.JS <Head/> component to each page with properties essential for enabling PWA. If adding new pages to the project, add the following <Head/> component to the HTML:

<Head>

```
<title>VisionAid</title>

<meta
  name='description'
  content='A nonprofit, advocating on behalf of persons with vision
  issues of any type'
/>
<meta name='theme-color' content='#ffffff' />

<link rel='icon' href='/favicon.ico' />

<link rel='apple-touch-icon' href='/apple-touch-icon.png' />

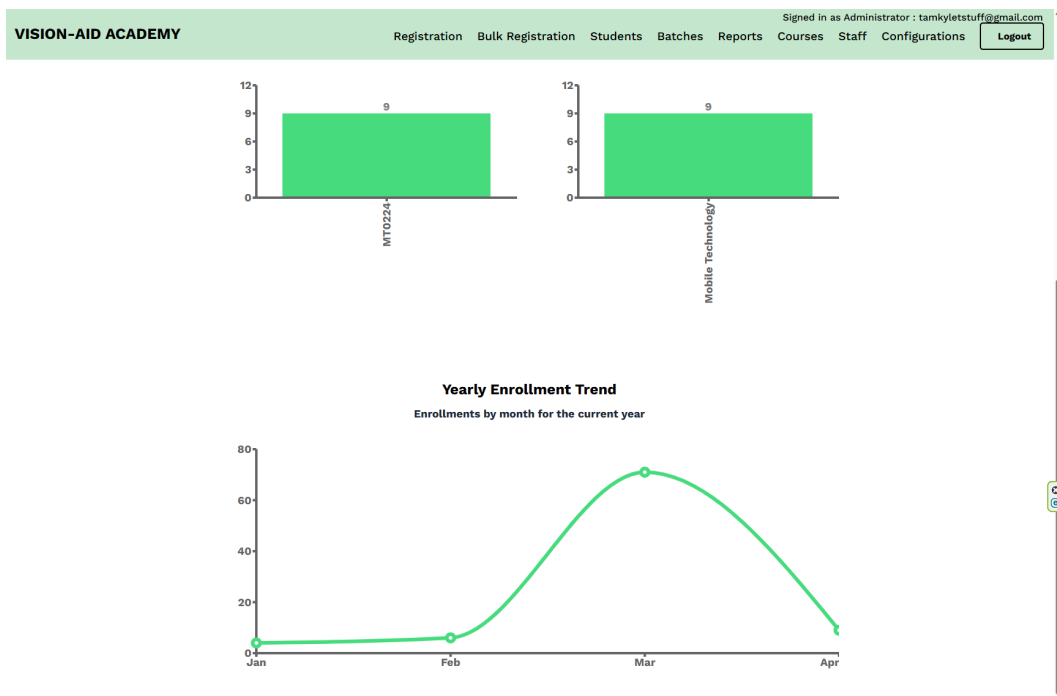
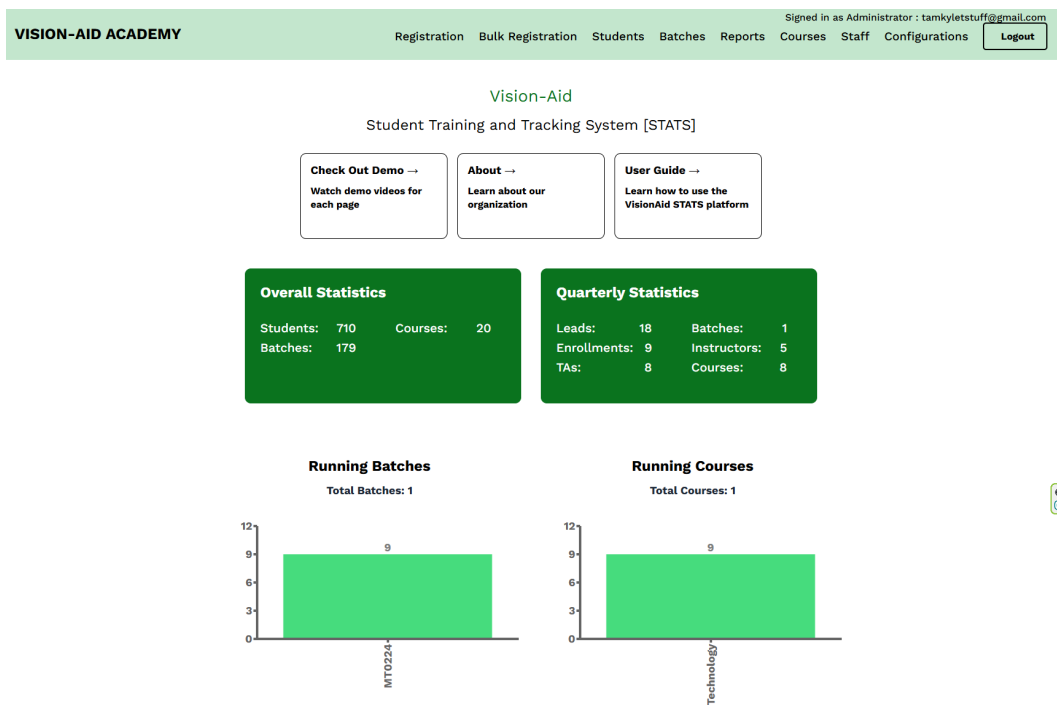
<link rel='manifest' href='/manifest.json' />
```

```
<link rel='preconnect' href='https://fonts.gstatic.com' crossOrigin />
<link
  rel='preload'
  as='style'
  href='https://fonts.googleapis.com/css2?family=Work+Sans:wght@400;500;700&display=swap'
/>
<link
  rel='stylesheet'
  href='https://fonts.googleapis.com/css2?family=Work+Sans:wght@400;500;700&display=swap'
  media='print'
  onLoad="this.media='all'"
/>
<noscript>
  <link
    rel='stylesheet'
    href='https://fonts.googleapis.com/css2?family=Work+Sans:wght@400;500;700&display=swap'
  />
</noscript>
</Head>
```

DELIVERABLE SCREENSHOTS

Dashboard Page

Overall statics, quarterly statistics, running batches chart, running courses chart, yearly enrollment trend graph, UX improvements (larger font, card formatting, layout, vertical bars and text for charts), landing page for students verses staff:



Bulk Registration Page

Registering many students simultaneously:

VISION-AID ACADEMY

RegistrationBulk RegistrationStudentsBatchesReportsCoursesStaffConfigurationsLogout

Signed in as Administrator : tamkyletstuff@gmail.com

Bulk Student Registration

+ Add Row

Check Rows

Register Valid Rows

Actions	Row ID	Row Status	Nam...	Gender	Date of Birth	Phone Number	Email	Parent / Guardian Phone Number	Country	State	City
	1	Not Yet Validated									

Privacy | Terms | Powered by Vercel

Students Page

Showing fields that can be editable (hovering shows pencil icon) and fixing fields that shouldn't be editable, sorting education field which can handle custom values:

VISION-AID ACADEMY

RegistrationBulk RegistrationStudentsBatchesReportsCoursesStaffConfigurationsLogout

Signed in as Administrator : tamkyletstuff@gmail.com

Student Management

Export to CSV

Actions	Id	Current Enrollment Status	Name	Education	Phone Number	Parent / Guardian Phone Number	City	State
	1022	Unassigned	Bulkregtest 3a	Below 10th Standard	4444444444		test	Badakhshan
	1019	Unassigned	Bulkreg Test01	Below 10th Standard	2342342341		test	Bengo Province
	1020	Unassigned	Bulkreg Test02	Below 10th Standard	1231232345		test	Andorra la Vella
	1014	Unassigned	Asdf2	Below 10th Standard	1234123444		test	Badakhshan
	1010	Unassigned	Asdf	Below 10th Standard	1234123444		test	Karnataka
	854	Enrolled	fd	Below 10th Standard	1234568900	1263647833	fdfd	Andhra Pradesh
	870	Dropout	Qweqwe	Below 10th Standard	1234567856	1234564456	Sn	Chhattisgarh
	806	Dropout	Brantley Patterson	Below 10th Standard	6784290267		Gillsville	Georgia
	1009	Enrolled	Test-aiish	Below 10th Standard	1367623762		ATL	Andaman and Nicobar Islands
	1008	Unassigned	Test Student 123	Below 10th Standard	1231231234		Test	Kalimantan Selatan
	904	Dropout	test13112	Below 10th Standard	4328943294		test13112	Andhra Pradesh
	850	Dropout	testing111111	Below 10th Standard	0	0	testing11111	Andhra Pradesh
	1007	Enrolled	Pooja Test 3	Below 10th Standard	6789056789		test	Jharkhand
	1006	Enrolled	Pooja Test1	Below 10th Standard	5678989898		test	Himachal Pradesh
	907	Enrolled	test432432	Below 10th Standard	1234555555		fdsfdfsd	Andaman and Nicobar Islands
	837	Enrolled	testtest1111111	Below 10th Standard	0		test	Arunachal Pradesh
	815	Unassigned	Test	Below 10th Standard	5555555555		Test	Maharashtra
	848	Enrolled	Example Student 2	Below 10th Standard	2345678901		Atlanta	Georgia
	857	Unassigned	testuser1	Below 10th Standard	5454354353	4382749832	testuser1	Assam
	858	Unassigned	phone test	Below 10th Standard	7328432783	1234567890	test	ANI

Batches Page

Add tooltip to show complete column name (for long names):

VISION-AID ACADEMY

RegistrationBulk RegistrationStudentsBatchesReportsCoursesStaffConfigurations

Signed in as Administrator : tamkyletstuff@gmail.com

Logout

Batch Management

Create New Batch

Actions	ID	Name	Batch Capacity	Enrolled Stud...	Dropped Stud...	Attendance %	Assessment %	Participation Cert...	Participati...	Passing Stud...	Pass %
121212	653	test course 11	100	2	0	0%	0%	0	0.0%	0	0.0%
1337	652	Computer for ...	0	8	0	0%	0%	0	0.0%	0	0.0%
MT0224	147	Mobile Techno...	0	9	5	50%	0%	4	44.4%	0	0.0%
777777789	639	C	2	16	0	25%	0%	2	12.5%	0	0.0%
000111222	640	C	5	1	1	0%	0%	0	0.0%	0	0.0%
1234564321	638	C	1	2	0	0%	0%	0	0.0%	0	0.0%
45678	643	Haki's Test Co...	3	2	0	42%	0%	0	0.0%	0	0.0%
12345678901234	645	C	3	3	1	0%	0%	0	0.0%	0	0.0%
123456654321	646	C	3	4	1	19%	0%	0	0.0%	0	0.0%
90909090	633	C	2	4	0	17%	0%	0	0.0%	0	0.0%
99999999	630	C	5	3	0	0%	0%	0	0.0%	0	0.0%
88888888	631	C	4	22	1	73%	0%	22	100.0%	0	0.0%
1234567890	634	C	1	1	0	0%	0%	0	0.0%	0	0.0%
7878787878	635	C	0	2	0	44%	0%	0	0.0%	0	0.0%
12344321	636	C	5	0	0	0%	0%	0	0%	0	0%
21342	649	Mobile Techno...	231	0	0	0%	0%	0	0%	0	0%
4448	623	HTML	0	4	0	100%	0%	4	100.0%	0	0.0%
9998	626	C	25	1	0	100%	0%	1	100.0%	0	0.0%
23329	624	CSS	0	0	0	0%	0%	0	0%	0	0%
9997	627	Mobile Techno...	25	3	0	3%	0%	0	0.0%	0	0.0%

Batch Management Page

Screen reader accessibility fix for when moving unassigned students to batches:

VISION-AID ACADEMY

RegistrationBulk RegistrationStudentsBatchesReportsCoursesStaffConfigurationsDashboard Settings

Signed in as Administrator : tamkyletstuff@gmail.com

Logout

Course: Mobile Technology, Batch: MT0224

Total students enrolled: 9

Batch AttendanceBatch GradesBatch StatusDocuments & FeesBatch ManagementAssessment Management

Assign Students to Batch

Show all students

Unassigned Students

ID	Name	First Choice	Second Choice
13	Papanaboina Shivaiah	HTML	Mobile Technology
132	Dilip Kumar	Mobile Technology	Python
411	Aakash Kumar	Mobile Technology	Python
806	Brantley Patterson	Mobile Technology	
816	Aaron Rodgers	Mobile Technology	Chatgpt
834	Kevin Taylor	Mobile Technology	C
870	Qweqwe	Mobile Technology	
959	Nabin Kim	asdfs	Mobile Technology
1018	Testperson For Bulkreg	Mobile Technology	

Current Batch

ID	Name	First Choice	Second Choice
481	Bharola Vijay Ramesh...		
540	Dannana Anitha		
505	Dhanavath Gopi		
321	Lakshya Garg		
541	Pandranki Jhansi		
542	Poodi Praveena		
504	Sandeep Rawat		
506	Shoukat Nabillal Gava...		
539	Surala Satyavathi		

Reports Page

Date zone fix (applies to the entire application), autocomplete or dropdown options in batchid or batchname for single batch report download, add filter for program manager:

VISION-AID ACADEMY

RegistrationBulk RegistrationStudentsBatchesReportsCoursesStaffConfigurationsLogout

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☒ Date Range Reports ☐ Single Batch Report ☐ Program Manager Reports

Date Range Batch Reports ⓘ

Select Date Range Type:

Custom Date Range

Start Date:

yyyy-mm-dd

End Date:

yyyy-mm-dd

Download Batch Report

Download Student Report



Courses Page

Sorting and filtering duration field, making duration field required:

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Course Management

+ New Course Form

Name	Description	Duration	Duration Type
C	c class	2	Months
Computer for good		1	Days
CSS	css class	12	Weeks
Haki's Test Course	This is a description	1	Weeks
HTML	Intermediate HTML	1	Months
Math 101	An into to math class	1	Months
Mobile Technology	To enable a person with visual impairment to effectively use the modern-day smartphone for day-to-day work, mobility and comfort	5	Weeks
PHP	a	2	Weeks
Python		7	Months
SEP B	Spoken English Programme Beginner	4	Weeks
test	test	5	Weeks
test	ds	7	Days
test 1	test 1	3	Weeks
test 4	eww	5	Weeks
test course 11	this is a test	3	Weeks
Test course III	this is a course test	6	Weeks
Test VP		5	Weeks

Staff Page

Edit cells fix, change some fields to uneditable:

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RegistrationBulk RegistrationStudentsBatchesReportsCoursesStaffConfigurationsLogout

All VisionAid Staff

+ New VA Staff

Name	Employee Number	Contact Number	Gender	Date of Birth	Visual Acuity	Work Location	Designation	Supervisor
Aishwarya	12345	3535343435332	F	14/3/1996	Sighted	Atlanta	Trainer	
Andrea Bogies	VSN597	87868	Female	15/5/2020	Sighted	Online	Teaching Assistant	
Arun Kumar	P006	8056018286	Male	16/8/1993	LowVision	Chennai	Teaching Assistant	
Babita Kumari Moharana	P060	9599353068	Female	21/7/1987	Blind	Andhra Pradesh	Program Manager	
Bhargav Vaghastiya	VSN585	9904494705		1/8/2000	Blind	Chennai	Training Coordinator	
Brantley Patterson	123	0	Male		Sighted	Atlanta	Teaching Assistant	Santosh Vemp
Brantley Patterson	BRP10	0		22/11/2024	Sighted	Atlanta	Telecaller	Santosh Vemp
Brett	VASTAT56001	7777777777	M	1/1/1995	LowVision	ATL	Training Coordinator	
C4G Spring 2026 Test Account	0987654321	0000000000	F	2/2/2000	LowVision	U.S.	Trainer	
CletVineetha Bosco	VSN8						Head of Training	
Cristina	VASTAT5603	1234567890	F	10/7/2000	LowVision	Atlanta	Trainer	
Divya Ramesh	GT5	4703013716	F	1/1/2000	Sighted	Atlanta	Program Coordinator	
divyatrainer	1	4703013716	F	3/9/2000	Sighted	Atl	Trainer	
ewfewf	656456	324324	M	26/2/2026	LowVision	fewfewf	Trainer	
Gajender Singh	P014	9509224972	Male	15/3/1999	Blind	New Delhi	MS Office Excel	Dhiraj Dutta V
Gurinder Singh	P042	7340967093	Male	19/7/2003	Blind	Bangalore	Trainer	Dhiraj Dutta V
Himanshi	P068	6239302502	Female	18/8/2001	Blind	Punjab	Teaching Assistant	Samira Silveri
Ishmeet Kaur	P016	8510999912	Female	25/5/1998	Sighted	New Delhi	Trainer	Dhiraj Dutta V
Justin McEllan	VSN595	1112223333		1/1/2000		Fort Worth	Trainer	
Kavya Thadakkamaila	610	+1 404509283			Sighted	Boston	Program Manager	Santosh Vemp

Configurations Page

Make dropdown fields configurable:

VISION-AID ACADEMY

RegistrationBulk RegistrationStudentsBatchesReportsCoursesStaffConfigurationsLogout

Signed in as Administrator : tamkyletstuff@gmail.com

Configurations

Dropdown values

Manage:Staff Designation

Current values

Head of Training

Delete

Program Coordinator

Delete

Program Manager

Delete

Teaching Assistant

Delete

Telecaller

Delete

Trainer

Delete

Training Coordinator

Delete

Add a new option

Value

e.g. Other

Add

Dashboard Settings Page

Toggle cards, charts, and graph:

VISION-AID ACADEMY

RegistrationBulk RegistrationStudentsBatchesReportsCoursesStaffConfigurationsDashboard SettingsLogout

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Dashboard Configurations

Dashboard section visibility

☒ Overall Statistics Card

☒ Quarterly Statistics Card

☒ Running Batches Chart

☒ Running Courses Chart

☒ Yearly Enrollment Trend Graph

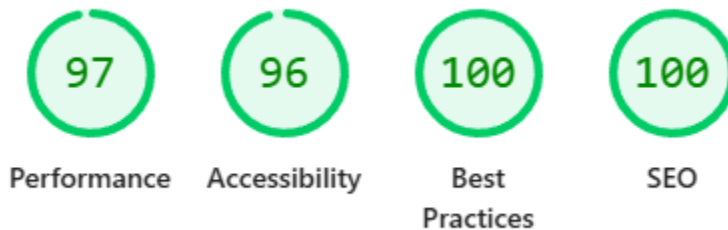
Save

Reset to Default

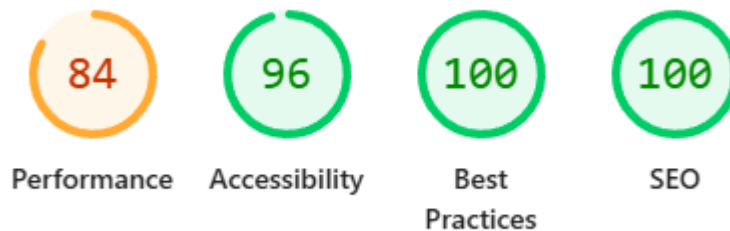


LIGHTHOUSE METRICS AND FORM FACTOR ANALYSIS

Desktop Lighthouse Metrics



Mobile Lighthouse Metrics



Portrait Form Factor Analysis

Responsive layout with vertical stacking of form fields, navigation menu collapses into a hamburger icon:



Vision-Aid

Student Training and Tracking System
[STATS]

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each page

About →

Learn about our
organization

User Guide →

Learn how to use the
VisionAid STATS platform

Overall Statistics

Students:	711	Courses:	20
Batches:	179		



Landscape Form Factor Analysis

Content fits better horizontally, input fields remain accessible and readable:

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Student Training and Tracking System [STATS]

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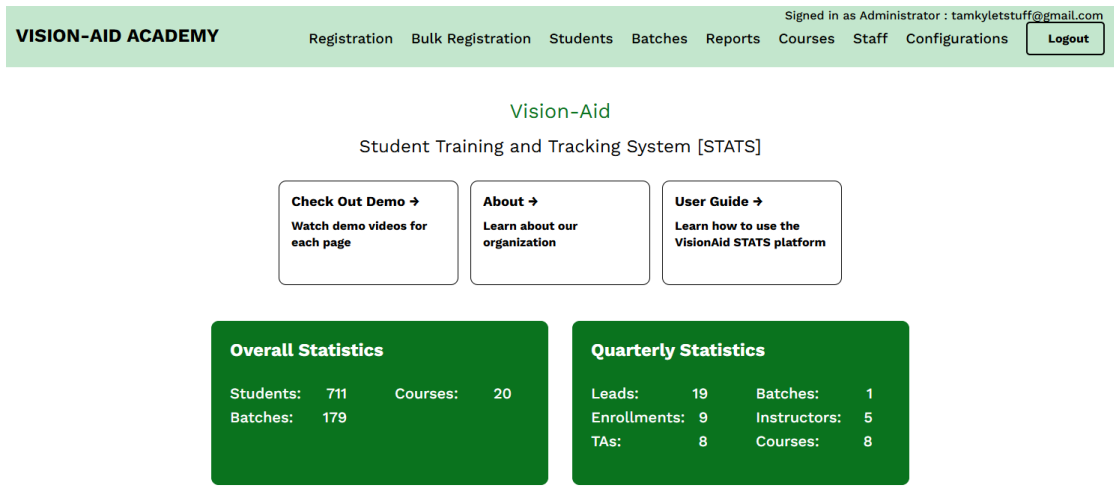
Learn how to use the
VisionAid STATS platform

Overall Statistics

Students:	711	Courses:	20
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Desktop Form Factor Analysis

Full layout loads with visible navigation and tables, editable tables and buttons work smoothly, best experience in terms of speed and visibility:



UX CONSIDERATIONS

- Stronger inline validation over alerts: several flows still use alert(...) (reports + form checks); replace with inline field errors/snackbars so users don't lose context.
- Duplicate detection UX: prefill by phone Date of Birth is great; add a visible "possible match" panel before submit so staff can choose update vs new registration confidently.
- Consistent success/failure messaging: standardize all feedback via GlobalSnackbar with predictable wording (Saved, Failed, Retry), including API failure reasons.
- Report builder usability: for reports, add presets ("This Quarter", "Last 30 days"), preview counts before download, and remember last used filters.
- Search/select for batch report: the dual batch name + ID fields can conflict; make one searchable combobox showing both (Name • ID • Course) and auto-fill the hidden id.
- Loading state quality: it already use overlays/spinners; add contextual loading text ("Fetching students...", "Generating CSV...") to reduce ambiguity.
- Accessibility consistency: we've done good work (skip links, keyboard handlers); next step is ensuring custom dropdown lists are fully keyboard + screen reader navigable.
- Mobile/tablet operations: AG Grid pages are dense; add responsive column profiles (essential fields first) and sticky action bar for export/create actions.
- Information hierarchy on dashboard: homepage charts are useful but visually dense; prioritize 3-5 KPIs first, then collapsible detailed charts.

STATEMENT OF KNOWN ISSUES / LIABILITIES

- Performance limitation: The website can take a long time to load certain datasets and pages, especially when processing larger volumes of records.
- Data entry liability: Current forms have limited input validation. Additional checks are needed to enforce consistent data formats and reduce invalid or inconsistent entries.

- Authentication constraint: Login currently depends on Google based authentication and organizational role mapping; access is effectively limited to approved accounts and may not support nonGoogle accounts without additional steps.
- Data quality dependency: Reports, dashboards, and derived batch/student status values depend on upstream data accuracy and consistent entry practices; incorrect source records can produce misleading outputs.
- Change-management risk: Ongoing configuration/rules updates (including batch status logic) can alter historical interpretation of records if governance and versioning are not strictly controlled.

PENDING FEATURE REQUESTS

- Multi-step student registration flow: Simplify the long registration form by splitting it into clear steps (Personal, Vision, Course, Context), with a visible progress indicator and auto-save draft support.
- Improved table layout and scrolling: Optimize the Student Management screen for easier scrolling, including better row density (e.g., 8-row viewport option) and reducing unnecessary spacing between the table and summary/data panels.
- Batch trainee export: Add the ability to download trainee details for a selected batch (e.g., CSV/Excel export).

PARTNER STATEMENTS

DONE (04/19/2026) by Divya	Partner understands developer documentation
DONE (04/19/2026) by Divya	Partner has gone through an installation walk-through